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Bihar Engineering University, Patna

B.Tech 2nd Semester Examination - 2026

Time: 03 Hours

Code: 100218 ***Subject: Python Programming*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option for the following questions (Answer any seven): **[2×7=14]**

(a) Python uses _____ to define blocks of code
- (i) Braces
- (ii) Indentation
- (iii) Parentheses
- (iv) Semicolon

(b) The operator used for exponentiation is
- (i) ^
- (ii) //
- (iii) **
- (iv) %

(c) The default step value in slicing is
- (i) 0
- (ii) 1
- (iii) -1
- (iv) 2

(d) Strings in Python are
- (i) Mutable
- (ii) Dynamic Arrays
- (iii) Immutable
- (iv) Numeric Objects

(e) Negative indexing begins from
- (i) First element
- (ii) Last element
- (iii) Middle element
- (iv) Zero index

(f) Dictionaries store data in the form of
- (i) Indexes
- (ii) Key-value pairs
- (iii) Lists
- (iv) Tuples

(g) The function used to combine two sequences into pairs is
- (i) join()
- (ii) pair()
- (iii) zip()
- (iv) merge()

(h) Pickle module is mainly used for
- (i) Image processing
- (ii) Networking
- (iii) Object serialization
- (iv) Graphics

(i) CSV stands for
- (i) Common Separated Values
- (ii) Character Separated Values
- (iii) Comma Separated Values
- (iv) Column Separated Variables

(j) Python is known as a _____ typed language.
- (i) Weakly
- (ii) Statically
- (iii) Dynamically
- (iv) Manually

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Q.2

(a) Explain the different features of Python Programming language. What are the different arithmetic operators in Python? Explain their precedence and associativity. [7]

(b) What are the basic data types in Python? How does Python decide the type of any variable? How a float is converted to an integer? Explain. [7]

Q.3

(a) Evaluate the importance of user-defined functions in Python. Compare them with built-in functions and discuss situations where each should be preferred. [7]

(b) Assess the significance of string slicing, indexing and formatting in Python. Demonstrate the output of the following statements and justify your answers. [7]

```
s = "Programming"
print(s[2:8])
print(s[-5:])
print(s[::-1])
```

Q.4

(a) Differentiate between Lists, Tuples and Sets on the basis of syntax, mutability, indexing, performance and applications. Support your comparison with suitable examples. [7]

(b) Construct a Python program that creates a dictionary containing employee ID, employee name and salary. Display all records and update the salary of a given employee using suitable dictionary methods. [7]

Q.5

(a) Write a python program to find Greatest Common Divisor of two numbers. [7]

(b) Illustrate with suitable examples the following list methods in Python: [7]
(i) insert (ii) append (iii) pop (iv) extend

Q.6

(a) Write the differences between the `if`, `if-else` and `if-elif-else` statements in Python. Provide code examples to demonstrate the usage and scenarios where each statement is preferred. [7]

(b) Explain the utility of `break` and `continue` statement with an example. [7]

Q.7

(a) What is dictionary and how it is created in Python? What is meant by key-value pairs in a dictionary? How does `del` operation work on dictionaries? Give an example. [7]

(b) Write a python program to read two numbers. Then find out whether the first number is a multiple of the second number. [7]

Q.8

(a) Design a Python program to create a text file, write the details of ten students into it, and subsequently read the stored data. Discuss the sequence of file operations involved in the program. [7]

(b) Justify the significance of the `os` and `os.path` modules in Python. Demonstrate the use of suitable functions to create directories, determine file paths and verify file existence. [7]

Q.9 Write Short notes on any two of the following: **[2×7=14]**

(a) Dynamic and Strongly Typed Language

(b) Type Conversion

(c) Scope and Lifetime of Variables

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(d) CSV File Handling

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